File number 1/2017

FINAL REPORT

INVESTIGATION ON AN ACCIDENT

THAI AIRASIA COMPANY LIMITED

AIRBUS A320-216, HS-ABB

AT KHON KAEN AIRPORT

ON 2 JANUARY 2017

THE AIRCRAFT ACCIDENT INVESTIGATION COMMITTEE

FOR THE AIRCRAFT ACCIDENT AND INCIDENT INVESTIGATION COMMISSION

THAILAND

This Final Report is produced for the purpose of obtaining information for enhancement of aviation safety in order to prevent same kind of accidents and serious incidents to happen again. It is not the purpose of this investigation to apportion blame or liability. It is not used for the police investigation, judicial or administrative proceedings aiming to blame or determine any persons to be punished in accordance with Section 64/7 and the last paragraph of Section 64/19 of the Air Navigation Act B.E. 2497 and an obligation under the Convention on International Civil Aviation done at Chicago on 7 December 1944 that Thailand is a signatory.

This document has been translated from Thai to English. In the event of any inconsistency between Thai and English versions, the original document in Thai shall prevail.

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ABBREVIATIONS

ACCREP	Accredited Representative
AFM	Aircraft Flight Manual
APU	Auxiliary Power Unit
ATC	Air Traffic Control
BEA	Bureau d'Enquêtes et d'Analyses pour la sécurité de l'aviation civile
CAAT	Civil Aviation Authority of Thailand
CAS	Calibrated Airspeed
CONF	Configuration
CRM	Crew Resource Management
CVR	Cockpit Voice Recorder
DCA	Department of Civil Aviation
DFDR	Digital Flight Data Recorder
FAF	Final Approach Fix
FCU	Flight Control Unit
FOM	Flight Operation Manual
ICAO	International Civil Aviation Organization
MSA	Minimum Sector Altitude
OM	Operation Manual
OML	Operational Multi-Pilot Limitation
PPC	Pilot Proficiency Check
RA	Radio Altimeter
RDAS	Repair Design Approval Sheet
RTAF	Royal Thai Air Force
STAR	Standard Arrival Route
SOP	Standard Operating Procedures
VLS	Lowest Selectable Speed
VOR	Very High Frequency Omni-Directional Range

SYNOPSIS

On 2 January 2017 at 05:42 p.m., an aircraft of Thai AirAsia Company Limited, Airbus A320-216, nationality and registration mark HS-ABB, flight FD3256, with 6 crew and 141 passengers, a total of 147 people on the flight route from Don Mueang International Airport to Khon Kaen Airport. While landing on runway 03 at Khon Kaen Airport, the aircraft had a tail strike. The aircraft sustained damage to the primary structure. There were no serious injuries or fatalities.

The Aircraft Accident Investigation Committee for the Aircraft Accident and Incident Investigation Commission Thailand notified the French Republic as the State of aircraft design and manufacture and the International Civil Aviation Organization.

The Aircraft Accident and Incident Investigation Commission Thailand conducted the investigation. The French Republic, as the State of aircraft design and manufacture had appointed accredited representative and advisers to participate in the investigation.

The Aircraft Accident Investigation Committee for the Aircraft Accident and Incident Investigation Commission Thailand made the Final Report of this investigation publicly available on 30 December 2025

Local time is used in this report. (UTC+7)

1. FACTUAL INFORMATION

1.1 History of the Flight

On 2 January 2017 at 05:42 p.m., a pilot and co-pilot operated an aircraft of Thai AirAsia Company Limited, Airbus A320-216, nationality and registration mark HS-ABB, flight FD3256, with 6 crew and 141 passengers, a total of 147 people from Don Mueang International Airport to Khon Kaen Airport. The pilot performed as Pilot Flying (PF) and the co-pilot performed as Pilot Monitoring (PM) during the landing on runway 03 using the Very High Frequency Omni-Directional Range (VOR) Y 03 KKC Approach Procedure (detail in Appendix 1). The weather at the airport was good, so the pilot attempted to practise Manual Approach and Landing without using the autopilot.

The pilot disengaged the autopilot at 10,000 feet and followed the Standard Arrival Route (STAR).

When the aircraft was at 3,400 feet, which was the minimum sector altitude (MSA) within 25 nautical miles of the airport, it was at the NISTA reporting point, which was the intermediate fix and intercept final course at 033 degrees, before descending to 2,200 feet.

When the aircraft reached the Final Approach Fix (FAF) point at 2,200 feet, with the landing gear down and the Landing Checklist completed, the pilot was cleared to land.

While the aircraft was at an altitude of 1,600 feet, after the control tower cleared to land, the pilot disengeged the Auto Thrust. The co-pilot who was performing as Pilot Monitoring reported an abnormality of the aircraft that changed from normal attitude while landing according to the Standard Operating Procedures (SOP).

The pilot corrected and controlled the aircraft to a normal glide slope, at which time the Precision Approach Path Indicator (PAPI) lights were showing three red lights and the wind direction was 045 degrees at 12 knots. The pilot corrected to a normal glide slope and reduced speed to 600 feet per minute.

At 100 feet, four red PAPI lights showed, and the co-pilot informed the pilot that the aircraft was below the glide slope.

At 30 feet, the pilot initiated a flare and immediately reduced engine power, causing a high sink rate. The aircraft was flying at a higher altitude and angle of attack higher than normal, resulting in a heavy impact with the runway.

Due to the higher than normal angle of attack, the aircraft had a tail strike. The aircraft sustained damage to the primary structure. There were no serious injuries or fatalities.

1.2 Injuries to Persons

Injuries to Persons	Crew	Passengers	Other
Fatal	-	-	-
Serious	-	-	-
Minor/None	-/6	-/141	-

1.3 Damage to Aircraft

The aircraft sustained significant damage to its lower fuselage at the rear, details as follows:

1.3.1 According to the Repair Design Approval Sheet (RDAS), the damaged is classified as a Major Repair (details in Appendix 4), as described below:

1.3.1.1 S19: lower skin flush repair.

1.3.1.2 S19: FR70 partial replacement lower section.

1.3.1.3 S19: pressure dome lower segment replacement.

1.3.1.4 S18: lower skin replacement.

1.3.1.5 Deviation during embodiment: replacement of two Hi-Lok fasteners (DAN 5-5-7) with blind-bolt (ASNA 0082).

1.3.2 External damage to the aircraft.

1.3.2.1 Scratch marks on the fuselage at Frame 67 – 71.

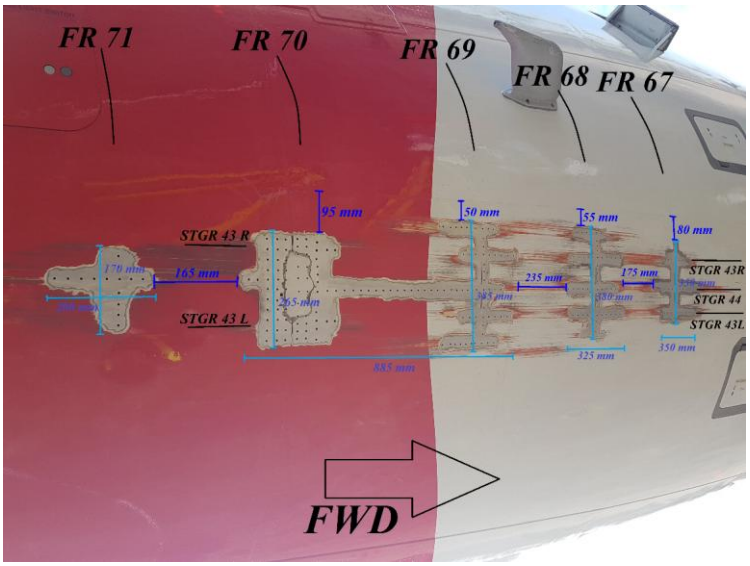


Figure 1: Damage to the lower fuselage at the rear of the aircraft.

1.3.2.2 Scratch marks on the lower fuselage at the rear of the aircraft.

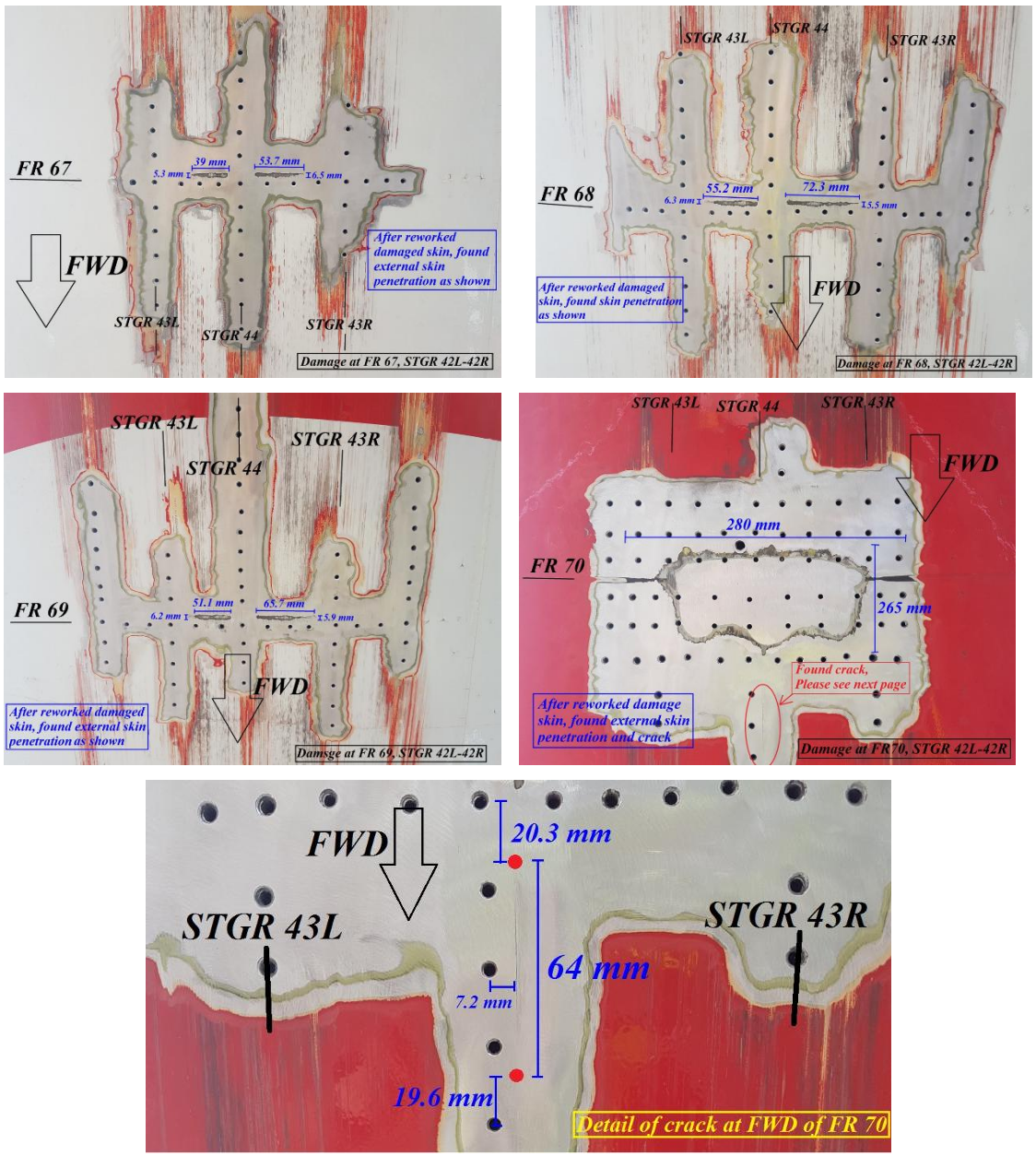


Figure 2: Damage to the fuselage.

1.3.2.3 Scratch marks on the Tail Cone, Drain Mast, APU Ram Air Inlet Flap and APU Drain Mast.

Figure 3: Damage to the APU Ram Air Inlet Flap and APU Drain Mast.

1.3.3 The Paint Flaking at Collar.

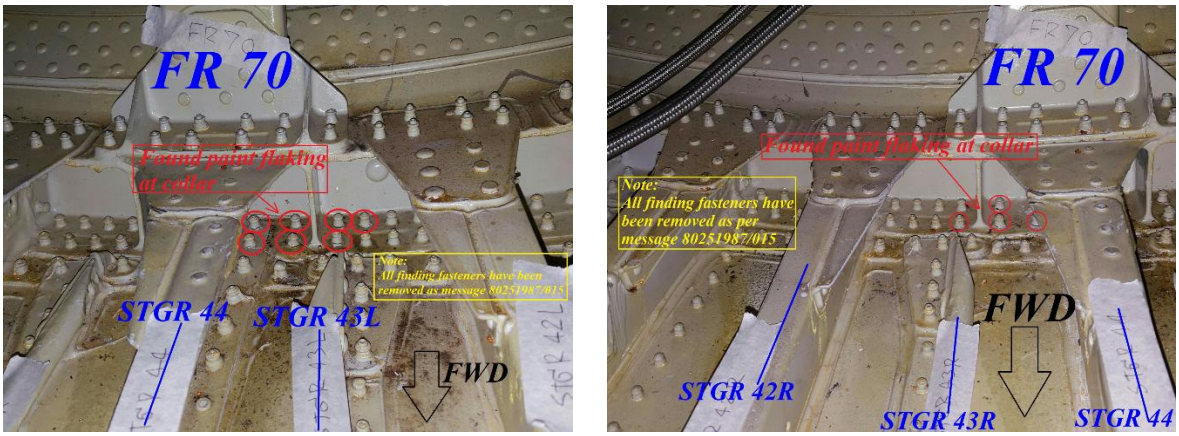


Figure 4: Damage to the inner fuselage at the rear.

1.4 Other Damage

None

1.5 Personnel Information

1.5.1 Pilot

Age:	36 years
Nationality:	Thai
Licence:	Airline Transport Pilot Licence was first issued by the Department of Civil Aviation (DCA) (Now, the Civil Aviation Authority of Thailand; CAAT) on 3 January 2013 and valid until 2 January 2018, with the following ratings: - Single & Multi engine land - Co-Pilot for Airbus A320 - PIC for Airbus A320
Medical certificate:	Class 1 was issued by Institute of Aviation Medicine, Royal Thai Air Force (RTAF) on 30 July 2016 and valid until 29 January 2017.
Medical limitation:	- Operational Multi-Pilot Limitation (OML) - Corrective lenses for distant vision are required on duty

Flying experiences before the accident are as follows:

- Total flight time	6,500:00	hours
- Flight time on A320	6,300:00	hours
- Flight time on A320 during the last 90 days	233:55	hours
- Flight time on A320 during the last 30 days	74:02	hours
- Last flight time on A320 on 1 January 2017	5:18	hours
- Rest period before the duty	17:54	hours

1.5.2 Co-Pilot

Age: 30 years

Nationality:	Thai
Licence:	Commercial Pilot Licence was first issued by DCA (Now, CAAT) on 30 June 2015 and valid until 29 June 2020, with the following ratings: - Single & Multi engine land - Co-Pilot for A320
Medical certificate:	Class 1 was issued by Institute of Aviation Medicine, RTAF on 4 June 2016 and valid until 3 June 2017.
Medical limitation:	Corrective lenses for distant vision are required on duty

Flying experiences before the accident are as follows:

- Total flight time	1,264:54	hours
- Flight time on A320	1,064:54	hours
- Flight time on A320 during the last 90 days	219:49	hours
- Flight time on A320 during the last 30 days	77:12	hours
- Last flight time on A320 on 1 January 2017	4:56	hours
- Rest period before the duty	25:07	hours

1.6 Aircraft Information

1.6.1 Aircraft History

1.6.1.1 Aircraft

Type:	A320-216
Manufacturing company:	Airbus S.A.S., The French Republic
Owner/ Possessor	Thai AirAsia Company Limited

Nationality and registration mark:	HS-ABB
Serial number:	3299
Maximum take-off mass:	73,500 Kilograms
Maximum landing mass:	66,000 Kilograms
Take-off mass on the accident flight:	60,512 Kilograms
Landing mass on the accident flight:	58,196 Kilograms
Certificate of registration:	Registered in the name of Thai AirAsia Company Limited, as the owner of the aircraft, received registration certificate number 6/2551 issued by DCA (Now, CAAT) on 10 March 2013.
Certificate of airworthiness:	Standard Certificate of Airworthiness 32/2558 was issued by CAAT, with validity from 8 November 2014 to 7 November 2017.
Time since New:	30,860:00 Hours

1.6.1.2 Left engine

Type:	CFM 56-5B6/3
Manufacturing company:	SNECMA, The French Republic
Serial number:	697618
Time since new:	28,288:00 Hours

1.6.1.3 Right engine

Type:	CFM 56-5B6/3

Manufacturing company:	SNECMA, The French Republic
Serial number:	697617
Time since new:	28,247:00 Hours

1.7 Meteorological Information

Aerodrome Routine Meteorological Report (METAR) of the Meteorological Unit at Khon Kaen Airport on 2 January 2017 were as follows:

1.7.1. At 04:00 p.m., surface wind direction 080 degrees, speed 7 knots, variable 040-120 degrees, visibility 10 kilometres or more, few clouds at 2,500 feet, temperature 31 degrees Celsius, dew point 19 degrees Celsius, atmospheric pressure above mean sea level 1012 millibars. There was no significant change within the next 2 hours.

1.7.2 At 05:00 p.m., surface wind direction 070 degrees, speed 5 knots, variable 040-120 degrees, visibility 10 kilometres or more, few clouds at 2,000 feet, temperature 30 degrees Celsius, dew point 19 degrees Celsius, atmospheric pressure above mean sea level 1013 millibars. There was no significant change within the next 2 hours.

1.7.3 At 06:00 p.m., surface wind direction 060 degrees, speed 5 knots, variable 040-120 degrees, visibility 10 kilometres or more, scattered clouds at 4,000 feet, temperature 28 degrees Celsius, dew point 19 degrees Celsius, atmospheric pressure above mean sea level 1014 millibars. There was no significant change within the next 2 hours.

1.8 Aids to Navigation

The navigation aids at Khon Kaen Airport functioned normally.

1.9 Communications

The communications between the pilot and the ATC at Khon Kaen Airport were normal.

1.10 Aerodrome Information

Khon Kaen Airport is located at latitude 16 degrees 27 minutes 52.39 seconds North and longitude 102 degrees 46 minutes 56.19 seconds East, 670 feet above mean sea level. Runway 03/21 has asphaltic concrete surfaces, and measures 3,050 metres long and 45 metres wide.

1.11 Flight recorders

1.11.1 The Flight Recorder (FDR) consists of a Digital Flight Data Recorder (DFDR), which records data for 26 hours before being overwritten. And the Cockpit Voice Recorder (CVR), which records data for 2 hours before being overwritten. The DFDR and CVR are mounted at the rear of the fuselage, where the air pressure is unpressurised. The CVR was not a significant factor to this accident.

1.11.2 The Aircraft Accident Investigation Committee for the Aircraft Accident and Incident Investigation Commission sent the DFDR to the Bureau d'Enquêtes et d'Analyses pour la sécurité de l'aviation civile (BEA), the DFDR could be read and analyse the data accurately and completely, detailed as follows:

Attitude	DFDR
10,000 feet	AP1 is voluntarily disengaged then the aircraft is manually handled from the CM1side.
1,600 feet	Once established on final segment around 1600ft RA, ATHR is disengaged. - Speed target 138 Knots (VLS +5 kt) - Slats/Flaps in CONFIG 3
500 feet	**The headwind progressively decreased from 12kt to ~0kt on ground.** - Pitch was progressively increased from +4.5° to +8° (reached at 100ft RA). - Thrust levers were slightly pushed forward.

Attitude	DFDR
350 feet	**CAS decayed below VAPP-5kt and reached 130kt (VAPP-8kt = VLS-3kt) around 80ft RA.** - Due to the speed decay, final approach was no more stabilised.
80 feet	**- Thrust levers were retarded to IDLE while CAS was 3kt below VLS.** - A nose down order was commanded, rate of descent increased up to 800ft/min and the aircraft went below the PAPI "4 reds".
50 feet	**A progressive nose up input was applied reaching full back stick at touchdown:** - The pitch gradually increased up to +10.5°. - As speed was decreasing well below VLS, the pitch up did not reduce the rate of descent : it remained around 700ft/min until touchdown. **The aircraft touched down hard, 50m before the runway threshold (on the displaced threshold) with:** +10.5° of pitch angle. -12ft/s (±2ft/s) of recalculated aircraft vertical speed. +2.5g of vertical load factor. CAS 123kt (VLS-10kt). **Nose up input was maintained during touchdown.** **Just after touchdown, the pitch continued to increase up to +13°:** **- The tailstrike occurred ~2 sec after touchdown.**

The results of the DFDR analysis from Airbus are summarised as follows:

At touchdown, pitch angle was high at +10.5°. The maintained nose up input during touchdown, made the pitch increase up to +13°, leading to a tailstrike 2 seconds after touchdown.

1.12 Wreckage and Impact Information

The aircraft touched down on the runway 50 meters before the runway 03 threshold. A scratch mark was visible on the runway, approximately 0.42 meters wide and 7.3 meters long, 22.9 meters from the runway's left edge line. The position where the aircraft touched down on runway 03 as shown in Figure 6.

Figure 5: Scratch mark on the runway.

Figure 6: The position where the aircraft touched down on runway 03.

1.13 Medical and Pathological Information

There were no pilot medical examination results after the accident occurred.

1.14 Fire

None

1.15 Survival Aspects

None

1.16 Test and Research

The Airbus Handling Quality analysis highlights the following contributing factors:

1.16.1 Hard landing and high pitch at touchdown:

1. The association of headwind decrease and commanded pitch increase without proper thrust adjustment led CAS to progressively decrease well below the speed target: from 350 feet RA the CAS was always below VLS, down to VLS-10kt at touchdown.

2. From 100 feet RA, the commanded nose down input increased the rate of descent up to 800 feet/minute but was not sufficient enough increase the speed.

3. From 50 feet RA, the pitch angle was increased up to +10.5°. Due to the speed decay down to VLS-10kt, the pitch up did not reduce the rate of descent leading the aircraft to suffered a hard landing at +2.5g (with ~-12feet/second of vertical speed).

1.16.2 Tailstrike:

1. At touchdown, pitch angle was high at +10.5°.

2. The maintained nose up input during touchdown, made the pitch increase up to +13°, leading to a tailstrike 2 seconds after touchdown.

1.17 Organisational and Management Information

None

1.18 Additional Information

1.18.1 Flight Crew Techniques Manual NO-130 - Non Precision Approach (P10-11/16) (Details in Appendix 5). A non-precision approach is a flight operation in which the aircraft is only controlled to remain in the centre of the glide path, rather than in the normal glide path. It is necessary to use visual reference during landing.

1.18.2 Standard Operating Procedures Manual SOP – Landing NOR-SOP-20 P4-6/8 (Details in Appendix 6) The topic of Landing specifies the conditions for Landing.

1.18.3 Flight Crew Operating Manual: LIM-Limitations LIM-12 - Environmental Envelope and Airport Operations P7/12 (Details in Appendix 7) specifies wind speed limits during landing, with tailwinds not exceeding 15 knots.

1.19 Useful or Effective Investigation Techniques

None

2 ANALYSIS

2.1 Flight operations

2.1.1 The pilot and co-pilot held valid licences with appropriate ratings and medical certificates.

2.1.2 The pilot may lack experience in Non Precision Approach landing.

2.1.3 The pilot attempted to practise manual approach and landing without using the autopilot. As a result, at 10,000 feet, the autopilot disengaged and the aircraft flew along the STAR.

2.1.4 At 1,600 feet, the pilot disengaged the Auto Thrust.

2.1.5 During landing, the aircraft flew below the normal glide path at a lower speed than the normal landing speed.

2.1.6 At 30 feet, the pilot initiated a flare and immediately reduced engine power, causing a high sink rate. The aircraft was flying at a higher altitude and angle of attack than usual, resulting in a tailstrike.

2.1.7 The Crew Resource Management (CRM) of both pilots were incomplete. The co-pilot did not warn the pilot to go around when the aircraft was in an unstabilised condition.

2.1.8 The landing speed did not exceed the limits specified in the flight operating manual.

2.1.9 According to DFDR, during the landing, the headwind speed decreased from 12 knots to 0 knots.

2.2 Aircraft

2.2.1 The aircraft had valid Certificate of Airworthiness and Certificate of Registration.

2.2.2 The aircraft's mass during landing did not exceed the value specified in the Operation Manual.

2.2.3 Aircraft Instrument performed normally.

2.2.4 All aircraft systems, including engines, were functioning normally. This aircraft did not have an automatic call out (Pitch Pitch) system.

2.3 Human Factors

2.3.1 Manual flying increases the pilot's workload, causing fatigue and decreasing decision-making efficiency in flight operations.

2.3.2 The pilot's overconfidence led to a lack of awareness of safety precautions (Lack of Awareness) (Complacency) and proper compliance to the flight operating manual.

2.3.3 There was no information regarding the medical examination following the accident.

2.4 Survivability

2.4.1 There were no serious injuries or fatalities from this accident.

2.4.2 This accident did not cause a fire.

3 CONCLUSIONS

3.1 Findings

3.1.1 The pilot and co-pilot performed a landing on runway 03 with VOR Y 03 KKC in Non Precision Approach.

3.1.2 The pilot disengaged the autopilot at 10,000 feet to practise flying manual approach.

3.1.3 The CRM of both pilots were incomplete.

3.1.4 During the landing, the headwind speed decreased from 12 knots to 0 knots.

3.1.5 The pilot performed the Landing Checklist before reaching the Final Approach Fix position at an altitude of 2,200 feet.

3.1.6 At 1,600 feet, the pilot disengaged the Auto Thrust.

3.1.7 The landing speed did not exceed the limits specified in the flight operating manual.

3.1.8 During landing, the aircraft flew below the normal glide path at a lower speed than the normal landing speed.

3.1.9 At 30 feet, the pilot initiated a flare and immediately reduced engine power, causing a high sink rate. The aircraft was flying at a higher altitude and angle of attack than normal, resulting in a tail strike.

3.2 Causes

This accident could have been caused by a rapid decrease in headwind, which increased the aircraft's sink rate, together with the pilot's late correction of the sink rate, which resulted in a tail strike.

3.3 Contributing Factors

3.3.1 Manual flying increases the pilot's workload, causing fatigue and decreasing decision-making efficiency in flight operations.

3.3.2 The pilot's overconfidence led to a lack of awareness of safety precautions and proper compliance to the flight operating manual. (Lack of Awareness , Complacency)

3.3.3 The co-pilot did not warn the pilot to go around when the aircraft was in an unstabilised approach.

3.3.4 The pilot lacked experience in manual flying.

4 SAFETY RECOMMENDATIONS

4.1 The Civil Aviation Authority of Thailand (CAAT) should inspect and supervise air operators in the following areas closely and on a regular basis:

4.1.1 Air operators should provide efficient and effective training to pilots, especially in the area of CRM.

4.1.2 Inspection of pilots capabilities, particularly on PPC, including emphasising flight operations in the case of non-precision approach landings, flight training without the use of autopilot. This includes the Stabilized Approach from 1,000 feet above ground level until touchdown, as well as performing a Go-around when required, which must be trained in a flight simulator.

4.1.3 Comply with the Air Navigation Act (B.E. 2497) and its amendments including other relevant laws, particularly Civil Aviation Board Regulation No. 67 regarding personnel on duty discipline, particularly item 7 (1) (Now the Civil Aviation Authority of Thailand Regulation No. 12, item 12(1)), must be followed, as well as the Aircraft Flight Manual (AFM) and Flight Operation Manual (FOM) related to Manual Flying for the aviation safety.

4.1.4 Conduct an inspection and evaluation of the air operator's safety management system (Conducting without the imposition of penalties) to determine whether hazards have been identified or reviewed, existing risk mitigations have been implemented, and additional risk mitigation measures have been established, such as additional training, testing, to ensure appropriate and effective safety management.

4.2 Air operators should

4.2.1 Improve flight operator training standards, taking into account position-specific knowledge and skills.

4.2.2 Strictly emphasise, review, and remind personnel on duty to comply with the Air Navigation Act (B.E. 2497) and its amendments including other relevant laws, particularly Civil Aviation Board Regulation No. 67 regarding personnel on duty

discipline, particularly item 7 (1) (Now the Civil Aviation Authority of Thailand Regulation No. 12, item 12(1)), must be followed, as well as the AFM and FOM related to Manual Flying.

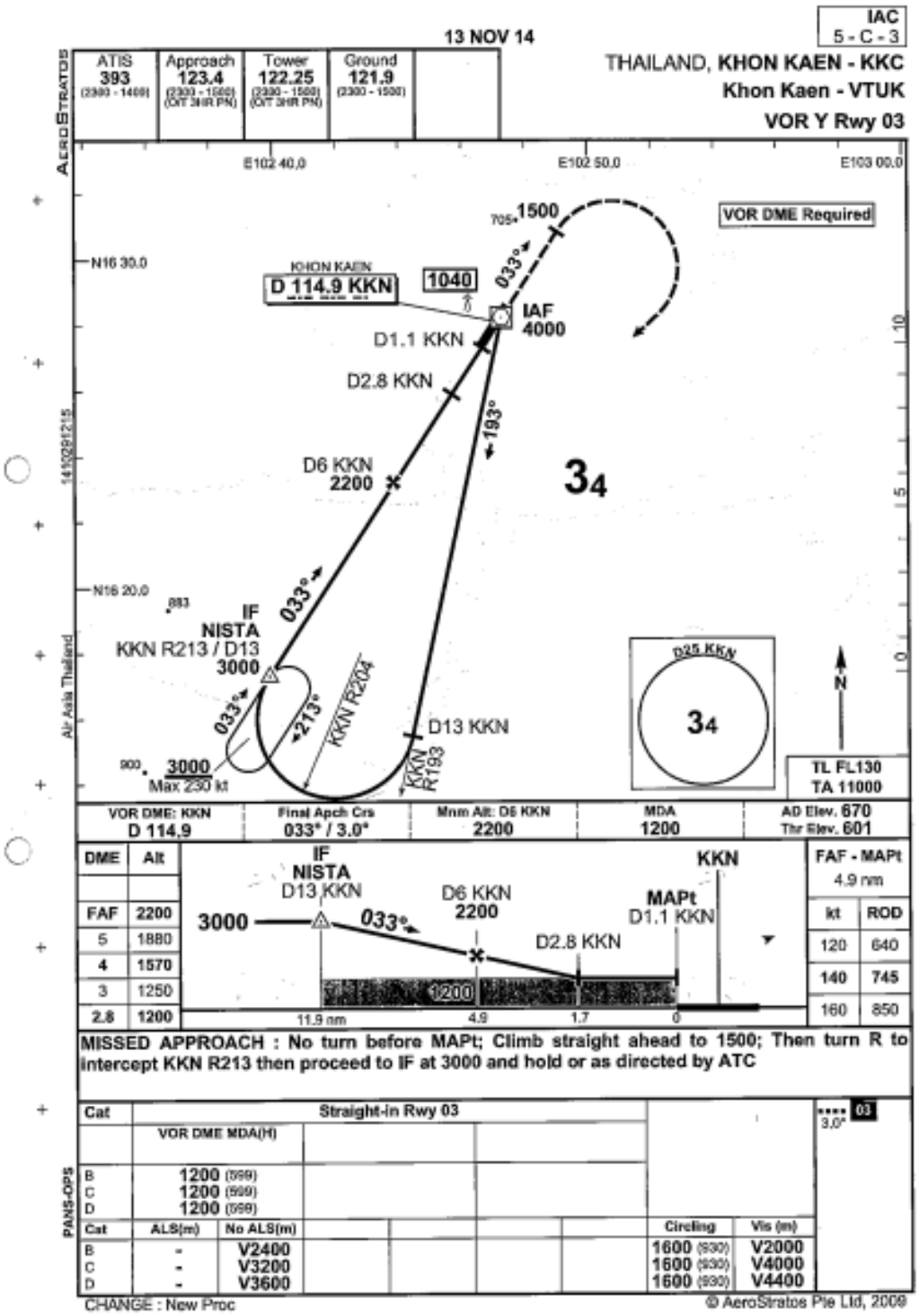
4.2.3 Emphasise review and remind pilots to strictly follow to the FOM and SOP, particularly the warning to go around when the aircraft is in an unstabilised approach.

4.2.4 Consider Emphasising on CRM among pilots during flight operations such as Leadership, Stress and Strain Management, Safety Culture, Situation Awareness, Monitoring and Intervention, Decision Making and Workload Management.

4.2.5 Consider improving training standards, surveillance systems, testing, and pilot knowledge and skills to be appropriate for their positions, particularly manual flying, including the Stabilized Approach from 1,000 feet above ground level until touchdown, as well as performing a Go-around when required, which must be trained in flight simulators. Additionally, conduct aviation safety with flight operators and other personnel involved in aviaition safety, focussing on areas such as PPC, additional training, testing, and warnings, to prevent similar accidents from occurring in the future.

APPENDIX 1

Instrument Landing Chart



APPENDIX 2

The DFDR analysis from Airbus Company

The DFDR analysis indicates the following key points:

- Autopilot1 and both FDs were voluntarily disengaged at FL100, and the whole approach was manually handled from the CM1 side.
- Once established on the final segment, at 1600ft RA, ATHR was disengaged, the speed target was selected at 138kt (VLS+5kt) with the slats/flaps in CONF3.
- The low parameters variations both on side sticks and accelerometers highlighted a calm approach.
- Below 500ft RA:
  - The headwind progressively decreased from 12kt to ~0kt on ground.
  - Pitch was progressively increased from +4.5° to +8° (reached at 100ft RA).
  - Thrust levers were slightly pushed forward.
- The association of headwind decrease and pitch increase without proper thrust adjustment led CAS to progressively decrease well below the speed target.
- From 350ft RA:
  - CAS decayed below VAPP-5kt and reached 130kt (VAPP-8kt = VLS-3kt) around 80ft RA.
  - Due to the speed decay, final approach was no more stabilized.
- Around 80ft RA:
  - Thrust levers were retarded to IDLE while CAS was 3kt below VLS.
  - A nose down order was commanded, rate of descent increased up to 800ft/min and the aircraft went below the PAPI "4 reds".
- Around 50ft RA, a progressive nose up input was applied reaching full back stick at touchdown:
  - The pitch gradually increased up to +10.5°.
  - As speed was decreasing well below VLS, the pitch up did not reduce the rate of descent : it remained around 700ft/min until touchdown
- The aircraft touched down hard, 50m before the runway threshold (on the displaced threshold) with:
  - +10.5° of pitch angle.
  - -12ft/s (±2ft/s) of recalculated aircraft vertical speed.
  - +2.5g of vertical load factor.
  - CAS 123kt (VLS-10kt).
- Nose up input was maintained during touchdown.
- Just after touchdown, the pitch continued to increase up to +13°:
  - The tailstrike occurred ~2 sec after touchdown.

APPENDIX 3

The Airbus Handling Quality

AIRBUS

The Airbus Handling Quality analysis highlights the following contributing factors:

Hard landing and high pitch at touchdown:

- ✓ The association of headwind decrease and commanded pitch increase without proper thrust adjustment led CAS to progressively decrease well below the speed target: from 350ft RA the CAS was always below VLS, down to VLS-10kt at touchdown.
- ✓ From 100ft RA, the commanded nose down input increased the rate of descent up to 800ft/min but was not sufficient enough increase the speed.
- ✓ From 50ft RA, the pitch angle was increased up to +10.5°. Due to the speed decay down to VLS-10kt, the pitch up did not reduce the rate of descent leading the aircraft to suffered a hard landing at +2.5g (with ~-12ft/s of vertical speed)

Tailstrike:

- ✓ At touchdown, pitch angle was high at +10.5°.
- ✓ The maintained nose up input during touchdown, made the pitch increase up to +13°, leading to a tailstrike 2 seconds after touchdown.

APPENDIX 4

Repair Design Approval Sheet (RDAS)

Repair Design Approval Sheet

1. RDAS Ref: 80251987/090/2017		**2. RDAS issue:** A	**3. Page 1 of 2**
4. A/C Type: A320-216 WV000-019	**5. MSN:** 03299	**6. ATA:** 53-41-00 53-51-00	**7. FC:** 21813 **FH:** 30862
8. P/N: –	**9. S/N:** –		

10. Title: TAIL STRIKE - SECTION 18 & 19 PERMANENT REPAIR

11. Damage Description:
Section S18 and S19 lower skin panel and internal structure get damaged due to tail runway contact.
For details see repair reference.

12. Repair Description:
- Section S19: lower skin flush repair
- Section S19: FR70 partial replacement lower section
- Section S19: pressure dome lower segment replacement
- Section S18: lower skin replacement
- Deviation during embodiment: replacement of two Hi-Lok fasteners (DAN 5-5-7) with blind-bolt (ASNA 0082) See next page(s)...

13. Regulations involved in addition to Certification Basis:
FAR Part26 Subpart E

14. Part 21A.449 Instructions for Continued Airworthiness[1]
Inspection areas : see details in Box 12

Threshold (or Inspection Starting Point)[2]:	
Interval:	
Method of Inspection:	
ALS Part 2 Inspection tasks adversely affected	YES ☐

[1]Only mandatory if classified as major in Box 16

[2]If not otherwise specified, threshold is from time of repair embodiment

15. Part 21A.443 Limitations

Repair Life Limitation[2][3]:	6 Months
Repair Structural Modification Point (SMP)[2]:	
Other Limitation:	
ALS Part 1 & Part 4 limitations adversely affected	YES ☐

[2]If not otherwise specified, limitations from time of repair embodiment

[3]If no repair category is indicated in Box 17, life limitation may be further extended after detailed analysis

16. Part 21A.435 Classification

☒ MAJOR

☐ MINOR

17. Repair Category

A ☐ Permanent repair with no additional requirements

B ☐ Permanent repair with inspection required

C ☐ Temporary or life limited repair

The technical information described above is approved under the authority of EASA approved Design Organisation Number EASA 21J031and per EASA rules Part 21 Subpart M. This approved data is based on the information provided by the requester to Airbus. Airbusdisclaims any and all responsibilities for incorrect or inaccurate information provided by the requester (including modification andSupplemental Type Certificate status). If this structural repair affects the compliance to a mandatory requirement, it is the operators'responsibility to obtain the necessary approval from its National Aviation Authority.

18. Designated Airworthiness Engineer Approval

Name: ANTOINE CABOS

Date: 16/03/2017

Signature and stamp: APPROVED UNDER EASA DESIGN ORGANISATION APPROVAL N° EASA.21.J.031 STAMP N° 318

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APPRNDIX 5

Flight Crew Techniques Manual

(Non Precision Manual)

A320 FLIGHT CREW TECHNIQUES MANUAL	NO – NORMAL OPERATIONS NO-130 – NON PRECISION APPROACH

FINAL APPROACH

Applicable to: ALL

It is essential that the crew does not modify the final approach in the MCDU FPLN page.

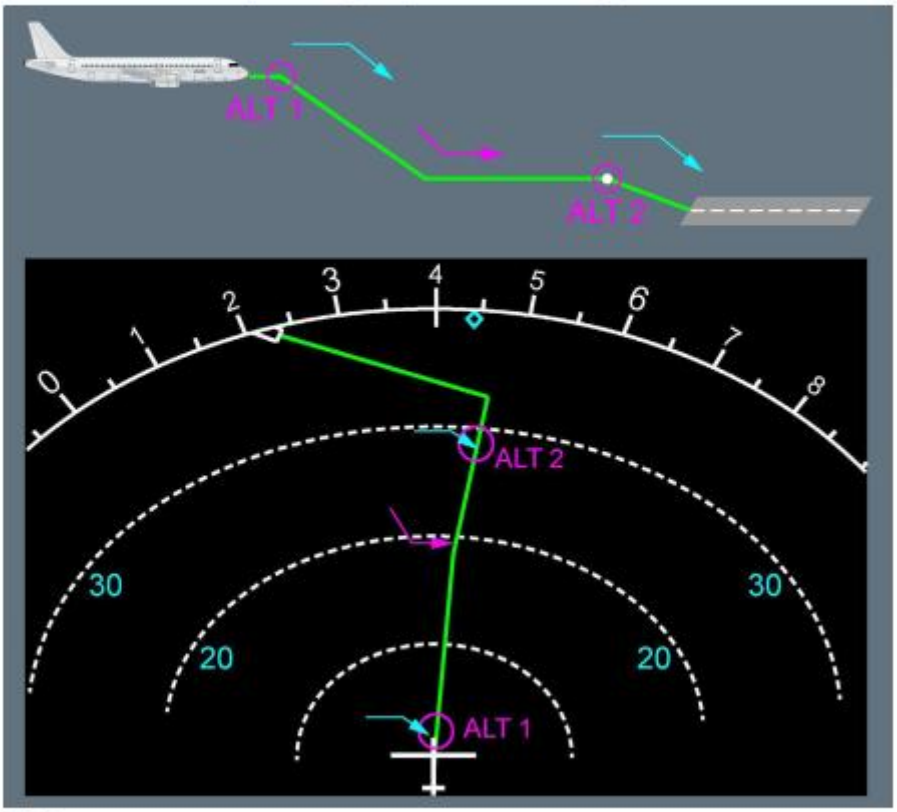
The final approach will be flown either
- Managed or
- Selected

MANAGED

For a managed approach, FINAL APP becomes active and the FMS manages both lateral and vertical guidance. The crew will monitor the final approach using
- Start of descent blue symbol on ND
- FMA on PFD
- VDEV, XTK, F-PLN on ND with GPS PRIMARY
- VDEV, XTK, F-PLN confirmed by needles, distance/altitude

If FINAL does not engage at the beginning of the final descent, the flight crew should consider to interrupt the instrument approach procedure unless they can maintain visual references throughout the approach.
In some NPAs, the final approach flies an "idle descent" segment from one altitude constraint to another, followed by a level segment. This is materialized by a magenta level off symbol on ND followed by a blue start of descent.

Final Approach Trajectory with Idle Descent Segment



SELECTED

For a selected approach, the Final Path Angle (FPA) should be preset on the FCU 1 NM prior to the the point where the final descent starts at the latest. A smooth interception of the final approach path can be achieved by pulling the FPA selector 0.3 NM prior to the point where the final descent starts.

APPENDIX 6

Standard Operating Procedures (Landing)

Thai AirAsia A320 STANDARD OPERATING PROCEDURES MANUAL	NORMAL PROCEDURES SOP – STANDARD OPERATING PROCEDURES SOP – Landing

LANDING

Applicable to: ALL

FLARE

The cockpit cut-off angle is 20 °.

● **In stabilized approach conditions, the flare height is approximately 30 ft:**

FLARE ..... PERFORM | PF
ATTITUDE ..... MONITOR | PM H209

"PITCH, PITCH" auto-callout is available on some aircraft and is activated when the aircraft pitch attitude is becoming excessive during flare and landing.

The PM should monitor the attitude, and call out: H210

- *"PITCH, PITCH", if the pitch angle reaches 10 °*
- *"BANK, BANK", if the bank angle reaches 7 °.*

THRUST levers ..... IDLE | PF

If autothrust is engaged, it automatically disconnects when the pilot sets both thrust levers to the IDLE detent.
In manual landing conditions, the "RETARD" callout is triggered at 20 ft Radio Altitude (RA), in order to remind the pilot to retard the thrust levers.

Note: The ground spoilers extension is inhibited if:

- *Both thrust levers remain above the idle detent, or*
- *One thrust lever is above idle and one thrust lever is at idle detent.*

GROUND CLEARANCE

Avoid flaring high.

A tailstrike occurs, if the pitch attitude exceeds 13.5 ° (11.7 ° with the landing gear compressed). H208

A wingtip or engine scrape occurs, if the roll angle exceeds 20 ° (16 ° with the landing gear compressed).

Be aware of the pitch-up tendency, with ground spoiler extension.

AT TOUCHDOWN

DEROTATION.....INITIATE | PF

- *Lower the nosewheel without undue delay.*
- *The PM continues to monitor the attitude.* H211

ALL THRUST LEVERS.....REV MAX or REV IDLE | PF

The flight crew must select reverse thrust immediately after main landing gear touchdown.

The flight crew must immediately select REV MAX, if any of the following occurs at any time during the landing:

- *An emergency*
- *The deceleration is not as expected*
- *A failure affects the landing performance*
- *A long flare or a long touchdown*
- *An unexpected tailwind.*

A small pitch up may occur during thrust reversers deployment before nose landing gear touchdown. However, the flight crew can easily control this pitch up.
As soon as the flight crew selects reverse thrust, they must perform a full-stop landing.

GROUND SPOILERS.....CHECK/ANNOUNCE | PM H212

Check that the WHEEL SD PAGE displays the ground spoilers extended after touchdown.

If no ground spoilers are extended:

- *Verify and confirm that both thrust levers are set to IDLE or REV detent.*
- *Set both thrust reverser levers to REV MAX, and fully press the brake pedals.*

Note: If ground spoilers are not armed, ground spoilers extend at reverser thrust selection.

REVERSERS.....CHECK/ANNOUNCE | PM H213

Check that the ECAM E/WD displays that the reverse deployment is as expected (REV green).

DIRECTIONAL CONTROL.....MONITOR/ENSURE | PF

- *Monitor directional control, if the rollout is automatic.*
- *Ensure directional control, if rollout is manual. Use rudder pedals for directional control.*
- *Do not use the nosewheel steering control handle before reaching taxi speed.*
- *During rollout, the flight crew should avoid sidestick inputs (either lateral or longitudinal).*
- *If directional control problems are encountered, the flight crew should reduce thrust to reverse idle until directional control is satisfactory.*

BRAKES.....AS RQRD | PF

- *Although the green hydraulic system supplies the braking system, if pedals are pressed rapidly, a brake pressure indication appears briefly on the BRAKE PRESS indicator.*
- *Braking may begin before the nosewheel has touched down, if required for performance reasons. However, when comfort is the priority, the flight crew should delay braking until the nosewheel has touched down.*

Note: *If no ground spoilers are extended, the autobrake is not activated.*

DECELERATION .....CHECK/ANNOUNCE | PM H214

The deceleration is felt by the flight crew, and confirmed by the speed trend on the PFD. The deceleration may also be confirmed by the DECEL light (if autobrake is ON).

AT 70 KT

SEVENTY KNOTS ..... ANNOUNCE | PM H215
BOTH THRUST LEVERS .....REV IDLE | PF

It is better to reduce thrust when passing 70 kt. However, high levels of reverse thrust may be used in order to control aircraft speed in the case of an emergency.

CAUTION	*Avoid the use of high levels of reverse thrust at low airspeed, unless required due to an emergency. The distortion of the airflow, caused by gases reentering the compressor, can cause engine stalls that may result in excessive EGT.*

APPENDIX 7

Flight Crew Operating LIM-Limitations LIM-12 – Environmental Envelope and Airport Operations

A320 FLIGHT CREW OPERATING MANUAL	LIM – LIMITATIONS LIM-12 – ENVIRONMENTAL ENVELOPE AND AIRPORT OPERATIONS

Applicable to: 9M-AFB to AFG, 9M-AFO, 9M-AFP, 9M-AFS to AFW, 9M-AFY, 9M-AFZ, 9M-AHA, 9M-AHB, 9M-AHD to AHH, 9M-AHJ, 9M-AHL, 9M-AHM, 9M-AHP to AHT, 9M-AHV, 9M-AHX to AHZ, 9M-AJA to AJE, 9M-AJG to AJK, 9M-AJN, 9M-AJP, 9M-AJS, 9M-AJT, 9M-AJV to AJZ, 9M-AQA to AQI, 9M-AQM to AQS, 9M-AQU, 9M-AQV, 9M-AQX to AQZ, D-AUBI, HS-ABA to ABZ, HS-BBA to BBW, MSN 7624, PK-AXC to AXJ, PK-AXR to AXV, PK-AXX, PK-AXY, PK-AZA, PK-AZD to AZJ, RP-C3227, RP-C3228, RP-C8189, RP-C8897, RP-C8970, RP-C8971, RP-C8972, RP-C8974, RP-C8975, RP-C8986, RP-C8987, RP-C8988, RP-C8992, RP-C8993, RP-C8994, RP-C8995, RP-C8996, VT-APJ, VT-ATB, VT-ATF, VT-BLR, VT-HYD, VT-JRT, VT-PNQ, VT-RED

Maximum demonstrated crosswind (takeoff and landing).....38 kt (gust included)

Note: - *The maximum demonstrated crosswind value is not an Airplane Flight Manual (AFM) limitation : It is the maximum crosswind condition experienced during the aircraft certification campaign.*
- *Airbus recommends that operators should not intentionally operate in crosswinds that exceed this value.*

Applicable to: 9M-AFB to AFG, 9M-AFO, 9M-AFP, 9M-AFS to AFW, 9M-AFY to AGB, 9M-AHA, 9M-AHB, 9M-AHD to AHH, 9M-AHJ, 9M-AHL, 9M-AHM, 9M-AHP to AHT, 9M-AHV, 9M-AHX to AHZ, 9M-AJA to AJE, 9M-AJG to AJK, 9M-AJN, 9M-AJP, 9M-AJS, 9M-AJT, 9M-AJV to AJZ, 9M-AQA to AQI, 9M-AQM to AQS, 9M-AQU, 9M-AQV, 9M-AQX to AQZ, 9M-NEO, D-AUBI, HS-ABA to ABZ, HS-BBA to BBZ, MSN 7591, MSN 7612, MSN 7624, PK-AXC to AXJ, PK-AXR to AXV, PK-AXX, PK-AXY, PK-AZA, PK-AZD to AZJ, RP-C3227, RP-C3228, RP-C8189, RP-C8970, RP-C8971, RP-C8972, RP-C8974, RP-C8986, RP-C8987, RP-C8995, RP-C8996, VT-APJ, VT-ATB, VT-ATF, VT-BLR, VT-HYD, VT-JRT, VT-PNQ, VT-RED

Maximum tailwind for takeoff ..... 15 kt

Applicable to: RP-C8897, RP-C8975, RP-C8988, RP-C8992, RP-C8993, RP-C8994

Maximum tailwind for takeoff ..... 10 kt

Applicable to: 9M-AFB to AFG, 9M-AFO, 9M-AFP, 9M-AFS to AFW, 9M-AFY to AGB, 9M-AHA, 9M-AHB, 9M-AHD to AHH, 9M-AHJ, 9M-AHL, 9M-AHM, 9M-AHP to AHT, 9M-AHV, 9M-AHX to AHZ, 9M-AJA to AJE, 9M-AJG to AJK, 9M-AJN, 9M-AJP, 9M-AJS, 9M-AJT, 9M-AJV to AJZ, 9M-AQA to AQI, 9M-AQM to AQS, 9M-AQU, 9M-AQV, 9M-AQX to AQZ, 9M-NEO, D-AUBI, HS-ABA to ABZ, HS-BBA to BBZ, MSN 7591, MSN 7612, MSN 7624, PK-AXC to AXJ, PK-AXR to AXV, PK-AXX, PK-AXY, PK-AZA, PK-AZD to AZJ, RP-C8189, RP-C8970, RP-C8971, RP-C8972, RP-C8974, RP-C8986, RP-C8996, VT-APJ, VT-ATB, VT-ATF, VT-BLR, VT-HYD, VT-JRT, VT-PNQ, VT-RED

Maximum tailwind for landing ..... 15 kt

Note: *For maximum tailwind for automatic landing and rollout, Refer to LIM-22-20 Maximum Wind Conditions for Cat II or Cat III Automatic Approach Landing and Roll Out.*

Applicable to: RP-C3227, RP-C3228, RP-C8897, RP-C8975, RP-C8987, RP-C8988, RP-C8992, RP-C8993, RP-C8994, RP-C8995

Maximum tailwind for landing ..... 10 kt